Joe Roe

I am a computational archaeologist using data science to understand the economy and ecology of the first farmers in Southwest Asia. I also develop research software and open data resources that support reproducible quantitative analysis in archaeology.

View this CV online with links at https://joeroe.io/cv

Academic appointments

Current | 2025

Assistant professor (*adjunkt*)

Department of Cross-Cultural and Regional Studies, University of Copenhagen

2024 | 2021

Research assistant (*Projektmitarbeiter*)

Institute of Archaeological Sciences, University of Bern

- Lead developer of XRONOS, an open repository for chronometric data (PIs: Dr. Martin Hinz, Prof. Albert Hafner)
- Collaborator on research project, "Climate Change Resilience and Vulnerabilities of Bronze Age Waterfront Communities (2200–800 BC)" (PI: Dr. Caroline Heitz)

2020 | 2017

Research assistant (*Videnskabelig assistent*)

Department of Cross-Cultural and Regional Studies, University of Copenhagen

- Responsibility for archaeological survey and spatial analysis components of major field projects in Iran and Jordan (PIs: Dr. Tobias Richter, Prof. Peder Mortensen)
- Developed databases and software interfaces for the projects' research data

Education

2025

PhD in prehistoric archaeology (*insigni cum laude*)

Institute of Archaeological Sciences, University of Bern

- Thesis: *Studies in open archaeology and the archaeoecology of early agriculture in West Asia, c. 15,000–6,500 Before Present*
- Supervisors: Prof. Albert Hafner, Dr. Martin Hinz

2017 | 2013

Postgraduate studies

Institute of Archaeology, University College London

- Computational modelling of prehistoric human ecology in the Southern Levant
- Supervisors: Prof. Louise Martin, Prof. Andrew Bevan, Dr. Andrew Garrard

Contact

joeroe@archaeo.dev
joeroe.io
0000-0002-1011-1244
Google Scholar
github.com/joeroe

Programming

- R
- SQL
- HTML/CSS
- Javascript
- Ruby
- Python

Made with pagedown (source)
Last updated 2026-09-27

2013 | 2012

MA Archaeology (Distinction)

Institute of Archaeology, University College London

- Dissertation: *Cultural phylogeny of kurgan burial mounds in the west Eurasian steppe*
- Supervisor: Prof. Stephen Shennan

2011 | 2008

BA Anthropology & Archaeology (1st class hons.)

Departments of Anthropology & Archaeology, Durham University

- Dissertation: *Spatial analysis of intra-site surface collection data from a Cucuteni-Tripolye mega-site*
- Supervisor: Prof. John Chapman

2008 | 2001

Secondary education

Woodkirk High School, Leeds

Publications

My publications reflect a number of ongoing collaborations with domain-specialist colleagues, bringing a data science perspective to addressing multidisciplinary research questions in prehistory.

2026

Roe, J., Yeomans, L., Arranz-Otaegui, A., Aldaia Aierbe, J., Proserpio, B., Jones, M., & Richter, T., Human Settlement and Water Availability at the Qa' Shubayqa during the Terminal Pleistocene and Early Holocene: Comparing Hydrological Models and Archaeological Data (Preprint). *EcoEvoRxiv* NA. https://doi.org/10.32942/X25T1T

Roe, J. & Arranz-Otaegui, A., Biogeography of crop progenitors and wild plant resources in the terminal Pleistocene and Early Holocene of West Asia, 14.7–8.3 ka. *Open Quaternary* 12. https://doi.org/10.5334/oq.163

2025

Roe, J., Schmid, C., Ebrahimiabareghi, S., Heitz, C. & Hinz, M., XRONOS: An Open Data Infrastructure for Archaeological Chronology. *Journal of Computer Applications in Archaeology* 8 (1): 242–263. https://doi.org/10.5334/jcaa.191

Martin, L., Roe, J. & Yeomans, L., Late Pleistocene-Holocene mammalian body size change in Jordan's Azraq Basin: A case for climate driven species distribution shifts. *Quaternary Science Reviews* 350: 109–147. https://doi.org/10.1016/j.quascirev.2024.109147

2024

Gaydarska, B., Roe, J., and Slavchev, V., Dietary Variability in the Varna Chalcolithic Cemeteries. *European Journal of Archaeology* 28 (1): 24–42. https://doi.org/10.1017/eaa.2024.33

Batist, Z. & Roe, J., Open archaeology, open source? Collaborative practices in an emerging community of archaeological software engineers. *Internet Archaeology* 67. https://doi.org/10.11141/ia.67.13

2023

Batist, Z. & Roe, J., open-archaeo: a resource for documenting archaeological software development practices. *Journal of Open Archaeology Data* 11: 9. https://doi.org/10.5334/joad.111

Arranz-Otaegui, A. & Roe, J., Revisiting the concept of the 'Neolithic Founder Crops' in southwest Asia. *Vegetation History and Archaeobotany* 32: 475–499. https://doi.org/10.1007/s00334-023-00917-1

2022

Arranz-Otaegui, A., Pedersen, P. N., Schmidt, A. F., Jörgensen-Lindahl, A., Roe, J., Villemoes, J., Pantos, G. A., & Killackey, K., Identifying the *chaîne opératoire* of club-rush (*Bolboschoenus glaucus* [Lam.] S.G.Sm) tuber exploitation during the Early Natufian in the Black Desert (northeastern Jordan). *Journal of Archaeological Science: Reports* 47. https://doi.org/10.1016/j.jasrep.2022.103677

Jones, M., Richter, T., Rollefson, G., Rowan, Y., Roe, J., Toms, P., Wasse, A., Ikram, H., Williams, M., AlShdaifat, A., Pedersen, P. N., & Esaid, W., The palaeoenvironmental potential of the eastern Jordanian desert basins (Qe'an). *Quaternary International* 635: 72–82. https://doi.org/10.1016/j.quaint.2021.06.023

Hinz, M., Roe, J., Laabs, J., Heitz, C., & Kolář, J., Bayesian inference of prehistoric population dynamics from multiple proxies: a case study from the North of the Swiss Alps. In NA, NA. NA, pp. NA. https://doi.org/10.31235/osf.io/dbcag

2021

Richter, T., Darabi, H., Alibaigi, S., Arranz-Otaegui, A., Bangsgaard, P., Khosravi, S., Maher, L., Mortensen, P., Pedersen, P., Roe, J., & Yeomans, L., The formation of Early Neolithic Communities in the Central Zagros: an 11,500 year old communal structure at Asiab. *Oxford Journal of Archaeology* 40 (1): 2–22. https://doi.org/10.1111/ojoa.12213

2020

Khosravi, S., Pedersen, P. N., Alibaigi, S., Roe, J., Heidary Dastenaei, M., Miladi, B., Darabi, H., Richter, T., & Mortensen, P., Archaeological Landscape of Razavar Valley in the Neolithic Period: the survey of the 8th–6th millennium BC in the north of Kermanshah Province. *Pazhoheshha-ye Bastan shenasi Iran* 26. https://doi.org/10.22084/NBSH.2020.21133.2094

Nebbia, M., & Roe, J., Landscape Studies. In Gaydarska, B., *Early Urbanism in Europe: The Trypillia Mega-Sites of the Ukrainian Forest-Steppe*. De Gruyter, pp. 60–107. https://doi.org/10.1515/9783110664959-007

Martin, L., & Roe, J., The Kheshiya Cattle Skull Ring: Zooarchaeological Analyses. In McCorriston, J., & Harrower, M. J., *Landscape History of Hadramawt: The Roots of Agriculture in Southern Arabia* (RASA) *Project*, 1998–2008. Cotsen Institute of Archaeology Press, pp. 275–348. https://doi.org/10.2307/j.ctvzgb8n3.20

2019

Henton, E., Roe, J., Martin, L., Garrard, A., Boles, O., Lewis, J., Thirlwall, M., & Jourdon, A-L., Epipalaeolithic and Neolithic gazelle hunting in the Badia of north-east Jordan. Reconstruction of seasonal movements of herds by stable isotope and dental microwear analyses. *Levant* 50 (2): 127-172. https://doi.org/10.1080/00758914.2019.1598764

Jones et al., 20,000 years of societal vulnerability and adaptation to climate change in southwest Asia. *WIREs Water* 6 (2): e1330. https://doi.org/10.1002/wat2.1330

2018

Arranz-Otaegui, A., González Carretero, L., Roe, J., & Richter, T., "Founder crops" v. wild plants: Assessing the plant-based diet of the last hunter-gatherers in southwest Asia. *Quaternary Science Reviews* 186: 263–283. https://doi.org/10.1016/j.quascirev.2018.02.011

2016

Martin, L., Edwards, Y., Roe, J., & Garrard, A., Faunal turnover in the Azraq Basin, eastern Jordan 28,000 to 9,000 cal BP, signalling climate change and human impact. *Quaternary Research* 86 (2): 200–219. https://doi.org/10.1016/j.yqres.2016.07.001

Publications (non-peer-reviewed)

2023 Roe, J., Peer Review Report For: Inglämnlagare: a tool for restructuring Swedish HER record site data for statistical analysis [version 2; peer review: 3 approved]. *F1000Research* 11 (1370). https://doi.org/10.5256/f1000research.157510.r222117

2016 Roe, J., PIA at 25: A Retrospective. *Papers from the Institute of Archaeology* 25 (2): p.Art. 16. https://doi.org/10.5334/pia.517

2015 Allentuck, A., Martin, L., & Roe, J., Refuse Disposal in the Early Epipalaeolithic? Preliminary Zooarchaeological Evidence from Kharaneh IV, Eastern Jordan. *Bulletin of the Council for British Research in the Levant* 10: 67–69. https://doi.org/10.1179/1752726015Z.00000000034

2014 Roe, J., Review of *Green Arabia*: Human Prehistory at the Crossroads of Continents. *Papers from the Institute of Archaeology* 24 (1). https://doi.org/10.5334/pia.458

Conference presentations

2026 Roe, J., *Can ecological models predict the occurrence of species in the archaeological record? Can I?*. Presented at the Computer Applications and Quantitative Methods in Archaeology (CAA) annual conference, Vienna, 31 March – 4th April 2026.

Roe, J., Batist, Z., Fellows Yates, J. A., & Titolo, A., *archaeo.social and archaeo.dev: decentralised, collective digital infrastructure for archaeologists*. Presented at the Computer Applications and Quantitative Methods in Archaeology (CAA) annual conference, Vienna, 31 March – 4th April 2026.

Roe, J. & Hinz, M., XRONOS: *An open data infrastructure for archaeological chronology*. Presented at the Computer Applications and Quantitative Methods in Archaeology (CAA) annual conference, Vienna, 31 March – 4th April 2026.

See my homepage for abstracts and posters.

2025

Heitz, C., Ismail-Meyer, K., Aellen, C., & Roe, J., *Leben am See in Zeiten des Klimawandels – Klimaarchäologische Perspektiven auf bronzezeitliche «Pfahlbauten» im zirkum-alpinen Raum*. Presented at the Berner Zirkel für Ur- und Frühgeschichte.

Hinz, M. & Roe, J., *Open-Source Bayesian Chronological Modeling: The Role of a Domain-Specific Language*. Presented at the Computer Applications and Quantitative Methods in Archaeology (CAA) annual conference, Athens, 5–9 May 2025.

Geeraerts, G., Roe, J., & Levy, E., *A command-line interface for chronological network modelling in R*. Presented at the Computer Applications and Quantitative Methods in Archaeology (CAA) annual conference, Athens, 5–9 May 2025.

Roe, J., *The chronoverse – towards a common language for chronological modelling in R*. Presented at the Computer Applications and Quantitative Methods in Archaeology (CAA) annual conference, Athens, 5–9 May 2025.

Richter, T. & Roe, J., *Using Ancient Environmental Genomics to Improve Our Understanding of Early Agro-Ecosystems in southwest Asia (and beyond)*. Presented at the 14th International Congress on the Archaeology of the Ancient Near East.

Martin, L., Shdeifat, A., Richter, T., Jones, M., Roe, J., Thaher, A., Alawamleh, A., Ibrahim, M., & Ghlelat, A., *Wadi al-Dahek: Report on the first season of archaeological survey and sedimentological assessment in Jordan's 'White Desert' in the eastern Badia*. Presented at the 16th International Conference on the History and Archaeology of Jordan (ICHAJ).

Richter, T. & Roe, J., *From Scatters to Structured Data: First Lessons from Analysing Large Lithic Datasets from Southwest Asia*. Presented at the 11th International Conference of the Pre-Pottery Neolithic Chipped & Ground Stone Industries of the Near East, Jalès, 11–14th November 2025.

2024

Heitz, C., Ismail-Meyer, K., Aellen, C., & Roe, J., *Coping with the Cold – Climate Change Resilience and Vulnerabilities of Bronze Age Communities during the 3.7 ka 'Löbben' Glacier Advance (ca. 1900–1450 BCE)*. Presented at Climate of the Past and Societal Responses to Environmental Changes (Climpast), Bern, 5–8 June 2024.

Roe, J. & Hinz, M., *XRONOS as a Resource for Archaeological Chronology in West Asia*. Presented at the Digital Ancient Near Eastern Studies (DANES) Conference, Virtual, 4–6 December 2024.

Heitz, C., Ismail-Meyer, K., Aellen, C., & Roe, J., *Climate Change Resilience and Vulnerabilities of Waterfront Communities – Mixed Methods Research on 'Pile Dwellings' around the Alps (4000-800 BCE)*. Presented at the 30th Annual Meeting of the European Association of Archaeologists (EAA), Rome, 28-31 August 2024.

Hinz, M. & Roe, J., *ESTER: Estimation of the prehistoric population of Eurasia based on a large number of records*. Presented at the World Neolithic Congress, Şanlıurfa, 4–8th November 2024.

Roe, J. & Hinz, M., *Estimating the prevalance of post-agricultural population declines through the global radiocarbon record*. Presented at the World Neolithic Congress, Şanlıurfa, 4–8th November 2024.

2023

Roe, J. & Hinz, M., XRONOS: *open chronometric data from archive to infrastructure*. Presented at Digital Archaeology Bern 2023.

Roe, J., *Modelling the range of wild plants and crop progenitors in the Late Epipalaeolithic–Early Neolithic Levant*. Presented at Digital Archaeology Bern – Ancient West Asia.

Roe, J., *More than just porridge: old materialism and the macroarchaeology of the Neolithic Revolution*. Presented at the 13th International Congress on the Archaeology of the Ancient Near East, Copenhagen.

2022

Hinz, M. & Roe, J., *Zoom and Enhance: high-resolution reconstruction of macro-scale processes with Bayesian hierarchical models*. Presented at the European Association of Archaeologists (EAA) Annual Meeting, Budapest, 31 August – 3 September.

Roe, J., *Are we what we eat? The Southwest Asian 'Neolithic Revolution' as macroarchaeological theory*. Presented at the European Association of Archaeologists (EAA) Annual Meeting, Budapest, 31 August – 3 September.

Roe, J. & Hinz, M., XRONOS: *challenges in building a global open repository for chronometric data*. Presented at the European Association of Archaeologists (EAA) Annual Meeting, Budapest, 31 August – 3 September.

2021

Roe, J., *Out with the new, in with the old: recent advances in palaeoecological modelling with open data*. Presented at the Association for Environmental Archaeology Spring Meeting 2021: Open Science Practices in Environmental Archaeology, Oxford.

Roe, J., *A revised radiocarbon chronology of Ganj Dareh*. Presented at the 53rd Annual Meeting of the Canadian Archaeological Association, 'Ganj Dareh Then and Now: Integrating Past and Current Research'.

Roe, J., *Nonequilibrium dynamics in models of human palaeoecology*. Presented at Computer Applications & Quantitative Methods in Archaeology (CAA), Limassol (Virtual).

Batist, Z. & Roe, J., *Open archaeology: a survey of collaborative software engineering in archaeological research*. Presented at Computer Applications & Quantitative Methods in Archaeology (CAA), Limassol (Virtual).

Roe, J., *Computational models of chronology and regional settlement in the Epipalaeolithic–Neolithic Zagros (20000–6000 BP)*. Presented at the 'Revisiting the Hilly Flanks: the Epipalaeolithic and Neolithic periods in the eastern Fertile Crescent' conference, Copenhagen.

Pedersen, P. N., Roe, J., & Richter, T., *Free foragers? A case for transient social inequality at Epipalaeolithic Shubayqa, eastern Jordan*. Presented at the 12th International Congress on the Archaeology of the Ancient Near East, Bologna.

Roe, J., *On uncertain ground: modelling human palaeoecology in the Azraq basin, eastern Jordan, 24,000–8,000 BP*. Presented at the 12th International Congress on the Archaeology of the Ancient Near East, Bologna.

2019

Roe, J., **Gorram shiny!* An accessible interface for reproducible landscape archaeology with `R` and `fieldwalkr`*. Presented at Computer Applications & Quantitative Methods in Archaeology (CAA), Kraków.

Richter, T., Arranz Otaegui, A., Jones, M., Roe, J., & Yeomans, L., *Environmental Crisis and Societal Collapse: Was the Younger Dryas a significant factor in triggering the transition from the Palaeolithic to the Neolithic in the Levant?*. Presented at 'Culture in Crisis: Flows of Peoples, Artifacts and Ideas', the 14th International Conference on the History and Archaeology of Jordan (ICHAJ 14), Florence.

Arranz-Otaegui, A., Roe, J., Pantos, G. A., Santana, J., Araus-Cabrera, J. L., Le Roux, P., & Richter, T., *Locally available or imported? Identifying the provenance of Natufian plant food and fuel resources at Shubayqa 1 (northeastern Jordan)*. Presented at the 18th conference of the International Workgroup for Palaeoethnobotany (IWGP), Lecce.

Buffington, A., Olson, K., Roe, J., Al Kathiri, A. A., McCorriston, J., *Over Mountains and Through Woods: survey results and preliminary spatial analysis of the 2018 field season of the Ancient Socio-Ecological Systems in Oman (ASOM) project*. Presented at the 53rd Seminar for Arabian Studies, Leiden.

Roe, J., & Richter, T., *Mutable mobilities: 15,000 years of nomadic settlement in the Qa' Shubayqa, eastern Jordan*. Presented at the 53rd Seminar for Arabian Studies, Leiden.

Pantos, G. A., & Roe, J., *A halfway house: open source tools for tidy photographic and 3D data management between the field and the archive*. Presented at Computer Applications & Quantitative Methods in Archaeology (CAA), Kraków.

Roe, J., *The Q Continuum: open source GIS tools for paperless field survey*. Presented at Computer Applications & Quantitative Methods in Archaeology (CAA), Kraków.

Roe, J., Pedersen, P. N., Khosravi, S., *A survey of early prehistoric sites in the Razavar Valley (Kermanshah, Iran)*. Presented at the British Association for Near Eastern Archaeology (BANEA) annual conference, Liverpool.

2018

Roe, J., *Is there anything R can't do?*. Presented at the Computer Applications and Quantitative Methods in Archaeology (CAA) conference, Tübingen.

Roe, J., *`fieldwalkr`: an R package for spatial sampling and field survey simulation*. Presented at the Landscape Archaeology Conference (LAC), Newcastle & Durham.

Richter, T., Arranz-Otaegui, A., Boaretto, E., Jones, M., Roe, J., & Yeomans, L., *Reconstructing Environmental Impacts on Late Pleistocene and Early Holocene Societies in the Marginal Zone of the southern Levant: A Case Study from Shubayqa*. Presented at the XVIII UISPP World Congress, Paris.

Roe, J., *The Zagros in Prehistory: geography, chronology, demography*. Presented at the British Association for Near Eastern Archaeology (BANEA) annual conference, Liverpool, 2019, and the KU Archaeology Day.

2017

Martin, L. & Roe, J., *The Complexities of Mammalian Body Size Change in Late Pleistocene to Holocene Southern Levant, 35 years on*. Presented at 'From the Mediterranean to the Atlantic: themes in European zooarchaeology', Museum of Natural History and Science, Lisbon.

2016

Roe, J., *A `greener` model of palaeoenvironments in the eastern Levant*. Presented at Unravelling the Palaeolithic, Southhampton.

2015

Allentuck, A., Martin, L., Roe, J., *Domestic maintenance practices in the Early Epipalaeolithic of the Jordanian steppe*. Presented at the 12th International Symposium of the ICAZ Archaeozoology of Southwest Asia and Adjacent Areas Working Group (ASWA), Groningen.

Roe, J. & Martin, L., *Open season: modelling prehistoric wildlife dynamics in Epipalaeolithic and Neolithic Jordan*. Presented at the British Association for Near Eastern Archaeology (BANEA) annual conference, London.

Allentuck, A., Roe, J., Martin, L., Macdonald, D., & Maher, L., *Zooarchaeological evidence for domestic maintenance in the Early Epipalaeolithic*. Presented at the British Association for Near Eastern Archaeology (BANEA) annual conference, London.

Software & datasets

2026 **controller: R package for controlled vocabularies**

2025 **fieldwalkr: R package for spatial sampling and survey simulation**

2024 **era: R package for years with an era**

2023 **rpaleoclim: R interface for paleoclimate data**

2023 **open-archaeo: a list of open source archaeological software and resources**

Zenodo

2020 **c14: R package for tidy radiocarbon data**

2020 **rintchron: R interface to IntChron**

2020 **stratigraphr: R package for archaeological stratigraphy and chronological sequences**

2020 **evoarchdata: datasets from studies of cultural evolution in archaeology**

2019 **Data from Uerpmann 1987, The Ancient Distribution of Ungulate Mammals in the Middle East, TAVO A27**

Zenodo

I am also an active contributor to a number of other open source software and open data projects.

2019 **Trypillia mega-sites of the Ukraine**
Archaeology Data Service

Field projects

2025 **Survey director**
Wadi al-Dahek Survey — Jordan

2019 **Field director**
Shubayqa Archaeological Project — Jordan

2018 **Field archaeologist**
Ancient Socioecological Systems in Oman (ASOM) project — Oman

2018 **Supervisor**
Razavar Valley Survey — Iran

2018 | 2015 **Field archaeologist & supervisor**
Shubayqa Archaeological Project — Jordan

2015 **Field archaeologist**
Kharaneh IV Project — Jordan

2014 | 2012 **Field archaeologist & supervisor**
Tripillia Mega-Sites Project — Ukraine

2012 **Supervisor**
Mirkovo Project — Bulgaria

2009 **Trainee**
Tripillia Mega-Sites Project — Ukraine

2007 **Trainee**
York Archaeological Trust — England

In the field, I specialise in site prospection, intrasite survey, digital recording and GIS.

I was responsible for the survey component Shubayqa Archaeological Project (2018–2019), including designing the research strategy, planning and obtaining permits, and directing teams on the ground.

Grants & scholarships

2022 **Young Researchers' Promotion Fund**

And other internal funds

University of Bern

- 7300 CHF for 'Digital Archaeology Bern 2023'
- Small grants for conference and workshop participation

2019 **Project grant**

Danmarks Frie Forskningsfond

- "Connecting the Dots: Reconfiguring the Indo-European family tree"
- Named applicant. PI: Dr. Thomas Olander

2018 **Fieldwork grant**

Det Danske Institut i Damaskus

- 115,000 DKK to support archaeological fieldwork in eastern Jordan (2019)
- Co-applicant with Dr. Tobias Richter

2018 **Fieldwork grant**

H. P. Hjerl Hansen Mindefondet for Dansk Palæstinaforskning

- 50,000 DKK to support archaeological fieldwork in eastern Jordan (2019)
- Co-applicant with Dr. Tobias Richter

2016 **Project grant**

C. L. Davids Fond og Samling

- "Tracking Cultural and Environmental Change: the Epipalaeolithic and Neolithic in the Seimarreh Valley, central Zagros"
- Named applicant. PIs: Prof. Peder Mortensen, Dr. Tobias Richter, Dr. Hojjat Darabi

2015 **Institute of Archaeology Award**

UCL Institute of Archaeology

London

- Individual grant (£1888) to participate in archaeological fieldwork in eastern Jordan

2014 **Research Students' Conference Grant**

London Centre for the Ancient Near East (LCANE)

2013 **PhD Studentship**

Leverhulme Trust

- Part of Research Project Grant RPG-2013-223

2012 **Research-preparation Masters Award**

Arts and Humanities Research Council

Teaching & public engagement

Promoting computational tools, and especially R, in archaeology has been a particular focus; I have experience in teaching at a range of levels, as well as informally mentoring students and colleagues.

2026 — **Invited speaker**
Writers Union of Najaf — Iraq

2024 — **Guest lecturer**
University of Copenhagen

- "Theory and Method in Near Eastern Archaeology" (CCRS)
- "Advanced Archaeological Methods" (Saxo Institute)

2024 | 2020 — **Instructor**
"R for Archaeologists" winter school — University of Pisa

- For five years I have led an intensive two-day workshop on data visualisation in R for archaeologists

2023 | 2022 — **Course lecturer**
Institute of Archaeological Sciences — University of Bern

- Lead instructor on BA-/MA-level courses GIS in Archaeology and Statistical Methods for Archaeological Data Analysis (SMADA)
- Recognised for "Excellence of Teaching" for SMADA 2023 (top 14% of courses in the Faculty of Humanities according to student satisfaction)

2021 | 2018 — **Course lecturer**
Department of Cross-Cultural and Regional Studies
University of Copenhagen

- BA: "Practical Archaeology 1" (Arkæologisk praksis 1, HNAB0100FU), "Practical Archaeology 2" (Arkæologisk praksis 2, HNAB0102EU), "Science and Theory of Science" (Videnskabsteori, HTOR0102FU)
- KA: "Theory and Method in Near Eastern Archaeology" (HNAK0101EU), "Environments and landscapes in the archaeological record of southwest Asia" (HNAB0102FU), "Neolithic to Early Urban Periods in the Near East" (HNAK0102EU), "Project-based course" (HNAK0100FU)

2020 — **Workshop organiser**
Department of Cross-Cultural and Regional Studies
University of Copenhagen

- Led an intensive two-day workshop on mobile GIS for linguistic field research

2016 | 2015 — **Postgraduate Teaching Assistant**
UCL Institute of Archaeology — University College London

- ARCL2028 Research & Presentation Skills

2016 | 2014 — **Tutor**
The Brilliant Club — Schools in London

- Designed and delivered a university-style tutorial course for primary and high school students

2015

Student Volunteer

Maths and Archaeology workshop — Petrie Museum of Egyptian Archaeology

- Outreach event for high school students organised by Dr. Alice Stevenson (Petrie Museum) and Dr. Luciano Rila (UCL Mathematics) and supported by the London Mathematical Society

Professional service

Peer review

Journals

- *Journal of Open Source Software*: #4335, #5811, #9260
- *Journal of Open Archaeology Data* (2)
- PCI Archaeology: #333, #712
- *Advances in Archaeological Practice* (1); *Open Archaeology* (1); *Open Quaternary* (1); *PLOS ONE* (1); *Journal of Archaeological Science* (1)

2026

Session organiser

CAA2026 — Vienna

- S16: *"All Models are Wrong": Learning from Failure in Computational Archaeology*

2023 | 2020

Member of Review College

CAA International

- Reviews completed: CAA2026 (3); CAA2025 (4); CAA2024 (4); CAA2023 (3); CAA2022 (7); CAA2021 (8); CAA2020 (3)

2023

Conference organiser

Digital Archaeology Bern 2023 — Bern

2021

Session organiser

CAA2021 — Limassol

- S17: *Tools for the Revolution: developing packages for scientific programming in archaeology?*

2019

Session organiser

CAA2019 — Kraków

- S06: *Recent advances in spatial statistics for archaeology*

2016 | 2014

Senior Editor

Papers from the Institute of Archaeology — University College London

2016 | 2014

Postgraduate Student Representative

UCL Institute of Archaeology — University College London

- Communications Committee
- Publications Committee

2014 | 2013

Editorial Assistant

Papers from the Institute of Archaeology

University College London

2014

Conference organiser

Postgraduate Zooarchaeology Forum (PZAF)

London

Professional memberships

current | 2026

START Centre for Sustainable Agrifood Systems

- "Biodiversity in future multifunctional landscapes" (BioLand) research hub

current | 2025

Managing Artificial Intelligence in Archaeology

- European COST Action 23141
- Management Committee Representative for Denmark

current | 2025

CAA Denmark / CAA Danmark

- Committee Member / Bestyrelsesmedlem

current | 2020

Society of Research Software Engineering

current | 2014

Council for British Research in the Levant

current | 2014

Computer Applications and Quantitative Methods in Archaeology International

- Member of Review College (2018–present)
- Member of 'Scientific Scripting Languages in Archaeology' special interest group (2019–present)

I am especially active in CAA International and its special interest group for scientific scripting languages.

Workshop & seminar participation

2026

"Desert archaeology and ecology"

Workshop

Al-Qadisiyah University

2026

"Chronological Modelling III: a Round Table on Time in Computational Archaeology"

CAA2026

Vienna

2024 **MOSAICchrono Summer School**
International Summer School — European Archaeological Centre, Bibracte
- Invited guest lecturer
- Led workshop on “radiocarbon and aoristic modelling”

2019 **“Scientific Scripting Languages in Archaeology” Roundtable**
CAA2019 — Kraków

2018 **“Connecting the Dots”**
International Summer School — Christian-Albrecht University of Kiel
- Development of the R package moin: Modeling Interaction in Landscape Archaeology

2017 **Shubayqa Project Workshop**
Workshop — University of Copenhagen

2016 **Near East Climate Workshop**
Workshop — University of Nottingham

Other employment & volunteering

current | 2010 **Editor & administrator**
English Wikipedia
- Community-appointed administrator
- Elected member of the Arbitration Committee (2019–2020)

2018 **Gateway Advisor**
Kirklees Citizens Advice Bureau — Dewsbury

2012 **Curatorial assistant**
Gradski Musej Vršac — Vršac

2012 **Software engineer**
TPP SystmOne — Leeds
- Front-end and data migration programming for a large clinical systems developer, working with Java and SQL

In my work with Wikipedia and other Wikimedia projects, I seek to promote engagement between academic archaeology and the free knowledge movement.